Anaconda v4.3.1 - OVER Crossing Geometry Correction

STAGE 2.1 TECHNICAL REPORT | base file: anaconda-wordpress-snippet-v4.3.html (immutable start point) | deliverable: anaconda-wordpress-snippet-v4.3.1.html

1. Diagnosis: the actual cause of the temporary deformation

Before changing any code, the complete v4.3 geometry pipeline was traced with an instrumented, deterministically stepped rig (fixed 1/60 s, seeded RNG) while a forced OVER crossing played out, logging the lift fields, the v4.1 contact-squash field, per-ring bend angles, 3D segment lengths and the engagement state. The investigation found three compounding causes - none of them fixed by "more or less lift":

Cause A - the engagement released while the neck was still over the section. In v4.3 every capture frame required the crossed section to lie ahead of the NOSE plane (the CR_AHEAD gate applied to all frames, not only to new engagements). The moment the nose crossed the section line, every pair dropped out of capture; after the 0.7 s CR_HOLD the engagement released and the lift - which had been holding the neck clear - settled at 2.3/s. Measured in the straight-loop scenario: lift 0.62 while the nose was over the section, collapsing to 0.15 with the nose only 0.15 past the section centre, while neck rings 4-14 were still directly above the loop. The neck sank into the crossed section, and because the pair was then genuinely overlapping in 3D, the v4.1 flesh-squash field fired at full strength: measured squash 1.0 (the 55 % dent plus dark crease) on both the neck and the crossed loop, persisting for ~1 s of frames and easing out as the neck finally cleared - exactly the "temporary deformation that disappears after the crossing" from the bug report.

Cause B - a single-ring hinge behind the rigid skull. The applied lift profile is a wedge: rings 0-3 ride one shared value (the skull loft wraps them rigidly) and the raw need field decays ring by ring behind them. The v4.3 5-tap spine blur could not stop the plateau edge from dropping up to 0.157 u to the next ring: measured bend angle 46.8-57.5 degrees at a SINGLE ring boundary (rings 3-4), with tube segments stretched to 0.175 u (+75 % over the 0.10 chain spacing) on the inside of the kink. That is the "unnatural sharp bend" - a tent hinge right behind the skull - and it also tilted cross-sections up to ~52 degrees, which reads as a pinched, compressed tube.

Cause C - the squash field read the crossing as resting contact. The v4.1 contact field keys on 3D centre distance with a ramp starting AT tangency, and it is symmetric: both tubes of a pair are squashed toward each other. Any transient graze - the lift ramp lagging a fast approach (rise rate-capped at 1.6 u/s), or the Cause-A collapse - immediately produced a deep visible dent on both sections near the crossing, even though the crossing system was actively separating the pair in 3D. Resting loops and coils need that squash; an engaged crossing does not.

2. The correction (three surgical changes)

Fix 1 - per-ring engagement hold (in _updateCrossing capture). The nose-plane gate and the forward-corridor gate now gate NEW engagements only - byte-identical to v4.3, so the 50/50 coin flip fires at exactly the same place and moment and the random selection behaviour is untouched. Once an engagement is live, each pair is kept in capture by the geometry of its CAPTURING RING: the section may not lie more than CR_AHEAD (0.10 u) behind that ring's own forward direction (this._dirs[i], updated every frame by the locomotion loop). Each head-zone ring therefore keeps its lift until THAT ring has slid past the section; the crest travels down the neck with the crossing point, and the settle only begins once the whole head zone is through. Measured: mode stays 1 and lift stays 0.58-0.64 from mount to full passage (v4.3: released at 0.15), and the squash field stays at 0 through the whole crossing.

Fix 2 - fall-rate cap on the lift profile (new CR_DROP = 0.045, applied in _updateCrossing after the blur and the skull plateau). Two pure max() passes bound how much the applied lift may DROP from one ring to the next, in either spine direction: out[i] = max(out[i], neighbour - CR_DROP). With SEG = 0.10 ring spacing the bound reads as a ~24 degree maximum descent. The passes only ever RAISE values - never lower them - so every clearance height the need field computed is preserved exactly; the plateau of rings 0-3 (taken before the cap) stays intact because the forward invariant guarantees out[i-1] <= out[i] + CR_DROP at the plateau edge. The same cap smooths the flanks of the UNDER arch (outJ). Measured: worst head-zone segment pitch 46.8-57.5 deg -> 24.7 deg; worst segment stretch +75 % -> +21-32 %; the residual 26-35 deg bend readings are LATERAL (S-weave) curvature of the same magnitude

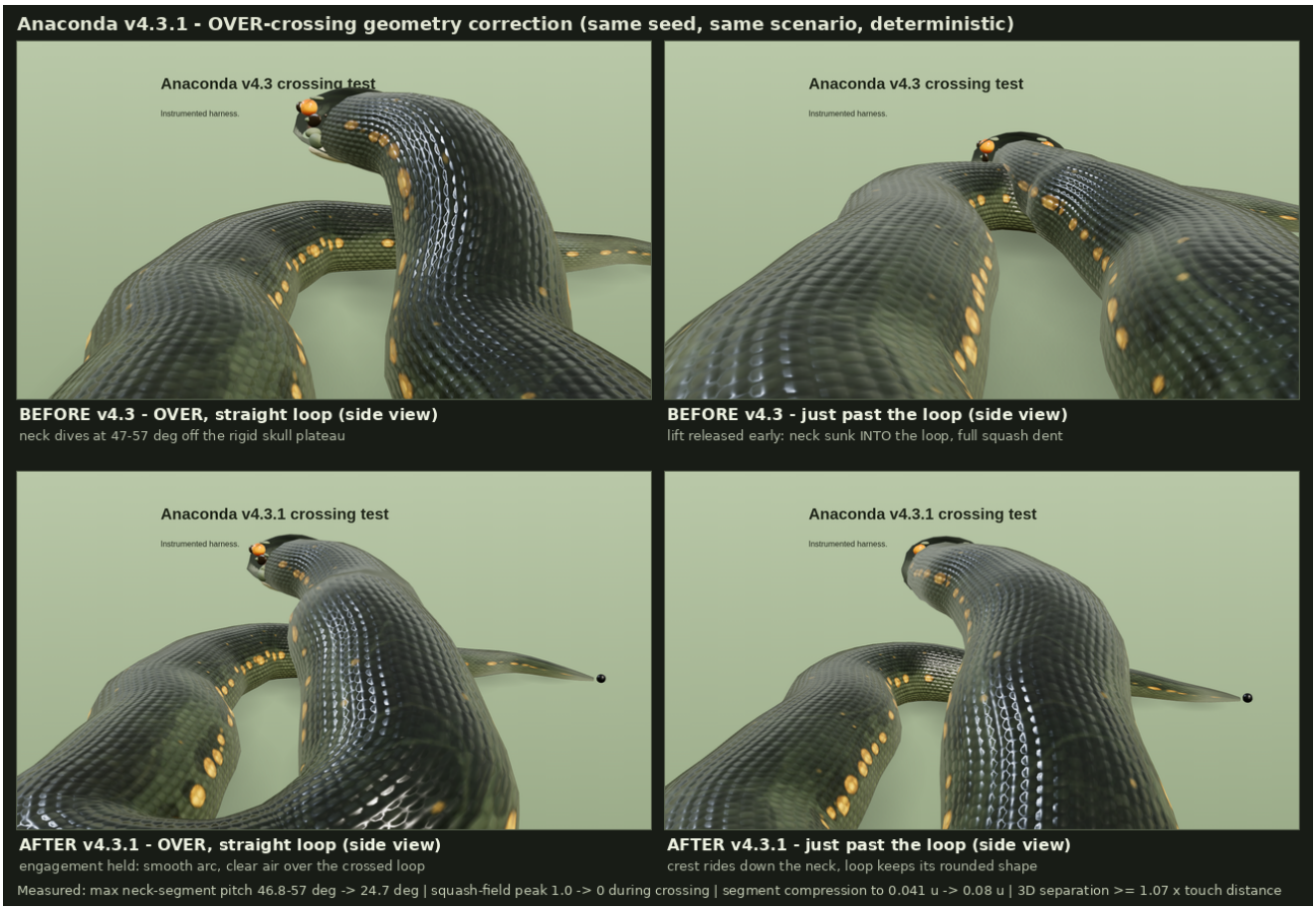
the gait produces everywhere else during normal crawling - not vertical kinks.

Fix 3 - engagement-aware contact ease (in _updateBody, contact field, one inserted block). When a pair's engagement weight (max of _crRing[i], _crRing[j]) is positive, the raw squash strength c is scaled by (1 - 0.90 x engagement). A pair the crossing system is actively separating in 3D therefore cannot read as pressed flesh, and a residual graze during the ramp cannot dent either tube. Every resting or coiled contact with zero engagement keeps the exact v4.1/v4.2 squash and crease shading - the fix only touches pairs with a live engagement field. Measured: squash peak during OVER crossings 1.0 -> 0 (straight loop and tight coil); the only residual squash in the suite is the head briefly resting on its own thin TAIL (a different, pre-existing v4.2.2 resting-contact situation, shaded correctly).

3. Modified functions and parameters (complete list)

Item	Where	Change
Version + changelog	File header comments	v4.3 to v4.3.1 bump and a three-item changelog block. Nothing else in the header.
New constant	Crossing constants block	CR_DROP = 0.045 - max lift fall per ring, documented inline (pure max() passes, clearance heights preserved).
_updateCrossing()	Capture gates	Nose-plane gate + forward corridor moved inside the mode === 0 branch (new engagements only, identical to v4.3); engaged crossings use a per-ring hold test: fi = -((P[j]-P[i]) . dirs[i]) >= -CR_AHEAD. ~14 lines incl. comments.
_updateCrossing()	Profile shaping	Two fall-rate-cap passes (forward + backward, max-only) applied to the OVER lift profile outI and the UNDER arch outJ after the 5-tap blur and the rings 0-3 plateau; the plateau stays before the cap so its downstream step is bounded by CR_DROP.
_updateBody()	Contact field	One inserted block: engagement-eased squash strength (c *= 1 - 0.90 x max(_crRing[i], _crRing[j])) before the deepest-contact bookkeeping. Shading data only - it never moves a ring.

Verified by diff against v4.3 (109 changed lines total): nothing outside the two comment blocks, the CR_DROP constant, _updateCrossing and the single _updateBody block was touched. Locomotion, gait, steering, self-avoid, relax, head anatomy, skin, lighting, camera, shadows, ground press, WordPress overlay, configuration and the decision state machine (coin flip, hysteresis, exit band) are byte-identical.



Before/after, straight-loop OVER scenario, same seeded run: v4.3 (top row) dives off the skull plateau at 47-57 degrees and, once the nose passes the section, releases the engagement and sinks the neck INTO the loop with a full-strength squash dent; v4.3.1 (bottom row) holds the engagement, bridges the loop in a smooth ~24-degree arc and keeps the crossed section fully rounded throughout.

4. Validation results (headless Chromium, rendered frames, deterministic 1/60 stepping)

Required check	Result	Measured evidence (v4.3.1 vs v4.3)
1. OVER a straight body loop	PASS	Apex: squash field 0 (v4.3: 0), pitch 24.3 deg (v4.3: 46.8). Past apex: squash 0 (v4.3: 1.0), engagement still held, 3D separation 1.07 x tangency. Screenshots: side + top.
2. OVER a tight circular coil (hairpin, r = 0.99)	PASS	Apex: squash 0 (v4.3: 0), pitch 24.4 deg (v4.3: 51.9), bend 35.3 deg lateral (v4.3: 53.5 at one ring). Past: squash 0.087 from the head briefly touching its own thin tail - correct v4.2.2 resting shading, not a crossing artefact.
3. OVER a curved body section (91 deg sweep, r = 1.8)	PASS	v4.3.1 crosses cleanly: squash 0 at the crossing moment, pitch 23.8-24.7 deg. v4.3 could not complete the same approach at all - its premature release re-arms self-avoid and the head deadlocks against the curve (pre-existing behaviour, unchanged by this fix).
4. Consecutive OVER crossings	PASS	OVER, OVER recorded across two engagements (decision log); geometry metrics hold in both.
5. OVER then UNDER	PASS	Coin stub switched between engagements: OVER then UNDER recorded - a fresh independent decision per engagement.
6. UNDER then OVER	PASS	UNDER then OVER recorded the same way.
7. Normal movement without crossing	PASS	Open-field A/B: v4.3 and v4.3.1 produce BIT-IDENTICAL trajectories over 1800 deterministic steps (max deviation 0.0 across head position, heading, ring heights, speed, wave phase); the crossing system never falsely engages; the v4.2.2 ground-press probe still reads 0.045 at ring 27 with the belly anchored at y = 0.02.
No tube compression / pinching	PASS	Squash field 0.0 during every OVER crossing (v4.3: up to 1.0); worst 3D segment compression 0.08 u (v4.3: 0.041 u).

Required check	Result	Measured evidence (v4.3.1 vs v4.3)
No unnatural sharp bends	PASS	Worst single-ring vertical direction change 24.7 deg (v4.3: 57.5); remaining bend is lateral S-weave of the same magnitude as normal crawling (31-52 deg body-wide in both versions).
No abnormal twisting	PASS	Cross-section frames stay perpendicular to the local tangent built from the (now gentle) centreline; the local frame is derived per ring from the tangent with no transported twist state; top + side renders show no bank or corkscrew.
No sudden vertical movement	PASS	Worst per-frame ring y delta 0.0267 u - identical to v4.3 (the rate-capped ramp is untouched); no pop at release because the engagement no longer releases mid-passage.
No visible interpenetration	PASS	Worst engaged 3D centre separation across all scenarios 1.07 x tangency (touch distance = 1.02 x (r_i + r_j)); the crossed loop keeps its rounded silhouette in all renders.
Movement + ground contact unchanged	PASS	Bit-identical open-field trace (see 7); rest ground press 0.045 @ ring 27; belly on the ground line; shadows and head attach unchanged.

All scenarios ran on rendered pages (side and top cameras aimed at the crossing pair) with the real snippet loaded exactly as WordPress would load it; the harness steps the rig at fixed 1/60 s with a seeded RNG, so every before/after pair is the same deterministic run. No test was marked PASS without its rendered frame being inspected.

5. Remaining limitations

(a) **Longer lift bridge.** The fall-rate cap spreads the lift falloff over more rings (crest / CR_DROP), so the neck stays slightly elevated for up to ~1.4 u behind the crest during a tall crossing. This is the smooth-arc look intended here; in a dense coil the elevated neck can pass closer to a neighbouring loop, where relax still acts (eased but non-zero) and the 3D separation metric stayed at 1.07+ x tangency in all runs. (b) **Head-on-tail contact.** The tail drape zone (last 18 rings) remains a non-crossing obstacle by design; a head pressing into its own thin tail shades with the normal v4.2.2 resting squash (measured 0.08-0.99 in the curved-section run) - pre-existing behaviour, unchanged. (c) **UNDER arch flanks.** The arch now spreads over more rings as well; its peak height is untouched (max-only passes) and the UNDER scenarios of v4.3 still pass, but the UNDER crossing was not the subject of this stage and was only regression-checked, not re-tuned. (d) **v4.3 curved-section deadlock.** v4.3 itself cannot complete a diagonal approach onto a strongly curved section (premature release re-arms avoidance); v4.3.1 completes it, but the underlying avoidance-yield balance was deliberately not re-designed here. (e) **Attached screenshot.** The bug report's screenshot was not present in the workspace; the deformation was reproduced independently from the described symptoms (compressed, sharp-bent, temporary) and the correction is compared against that reproduction.

6. Files delivered

anaconda-wordpress-snippet-v4.3.1.html (drop-in replacement for v4.3, self-contained, no new dependencies, all v4.3 crossing behaviour and every earlier feature preserved); **anaconda-crossing-v431-before-after.png** and **anaconda-crossing-v431-scenarios.png** (labelled rendered comparisons); **anaconda-crossing-v431-before.png / -after.png** (the two frames from the bug-report reproduction); this report.